

Practice Tutorial 1 - Python

Note: this tutorial is a required assignment and must be completed on a pass/fail basis.

A passing grade is needed in each practice tutorial to complete the course.

Please change the file name to: **tutorial1-ID1-ID2** e.g. tutorial1-123456789-987654321.

Your tutorials are checked *automatically*, ensure the filename is written *EXACTLY* as specified above.

Assignment #1

Write a function named **cube_volume** that when given a length of a line, returns the area of a cube.

For example cube_volume(7) returns 343.

```
In [5]: def cube_volume(len):
        if(len<=0):
            print('ERROR please enter a positive number!')
            return
        return (len**3)

        #we assumed that you can input any length and it prints an error if <0
        cube_len = int(input('Please enter a cube len'))
        print(f'the cube volume : {cube_volume(cube_len)}')

        #print(f'the cube volume : {cube_volume(7)}')
```

the cube volume : 343

the cube volume : 343

Assignment #2

Write a function named **parse_words**, that given a string prints each word in a different line.

For example parse_words ("***Lorem ipsum dolor sit amet, vis dolore soleat***") will be printed as

Lorem

ipsum

dolor

sit

amet,

vis

dolore

soleat

```
In [15]: def parse_words(lines):  
    if lines:  
        names_list = lines.split()  
        for s in names_list:  
            print(f'{s} \n')  
    else :  
        print("string is empty")  
    return  
  
    str_line = input('Please enter a string:')  
  
    print('\n')  
    parse_words(str_line)
```

Lorem

ipsum

dolor

sit

amet,

vis

dolore

soleat

Assignment #3

```
In [23]: state_names = ['Alabama', 'Alaska', 'Arizona', 'Arkansas', 'California', 'Colorado',  
                        'Delaware', 'Florida', 'Georgia', 'Hawaii', 'Idaho', 'Illinois', 'In',  
                        'Kansas', 'Kentucky', 'Louisiana', 'Maine', 'Maryland', 'Massachuset',  
                        'Minnesota', 'Mississippi', 'Missouri', 'Montana', 'Nebraska', 'Neva',  
                        'New Jersey', 'New Mexico', 'New York', 'North Carolina', 'North Dak',  
                        'Oklahoma', 'Oregon', 'Pennsylvania', 'Rhode Island', 'South Carolin',  
                        'Tennessee', 'Texas', 'Utah', 'Vermont', 'Virginia', 'Washington', '  
                        'Wisconsin', 'Wyoming']  
  
    # it needs to be defined before using update  
    state_name_length = {}
```

```

for state in state_names:
    state_name_length.update({state: len(state)})

print(state_name_length.get("Alaska"))
print(state_name_length.get("New York"))

```

6
8

The *list* **state_names** contains all names of the states that comprise the United States.

Create a *dictionary* data structure, called **state_name_length** such that when a valid state name is entered as a key the length of the name of the state will be returned as its value.

- example 1: `print(state_name_length.get("Alaska"))` will print 6.
- example 2: `print(state_name_length.get("New York"))` will print 8.

Assignment #4

```

In [30]: #creating a list of 1000 random integers, with each integer
#between 1 and 1000.
from random import randrange

a_list = []
for i in range(1,1001):
    x = randrange(1,1001)
    a_list.append(x)

```

4.1 Create a data structure

Create a *set* data structure named **a_set** from **a_list**. Find and print the length of the **a_set**.

```

In [38]: a_set=set()
for i in a_list:
    a_set.add(i)

print(f'len of the set :{len(a_set)} \nwhile len of the a_list :{len(a_list)}')

```

```

len of the set :636
while len of the a_list :1000

```

4.2 Practice the use of Markdown

Explain using the chunk below and with **Markdown** , why **a_set** has a smaller length compared to **a_list** (add a markdown block and write your answer in it).

why **a_set** has a smaller length than **a_list**?

in a set there is no duplicates (doesn't save duplicates) so in a list saves all numbers as well as duplicates

Assignment #5

Write a function **today_date** that prints today's date in the format DD-MM-YYYY.

For example if today is March 10, 2023, the function will print *10-03-2023* .

```
In [55]: from datetime import datetime

def today_date():
    current_date = datetime.now()
    Day= str(current_date.day).zfill(2)
    Month =str(current_date.month).zfill(2)
    Year=str(current_date.year)
    print(f'Today is: {Day}-{Month}-{Year}')
    return

today_date()
```

Today is: 24-03-2025

```
In [ ]:
```